

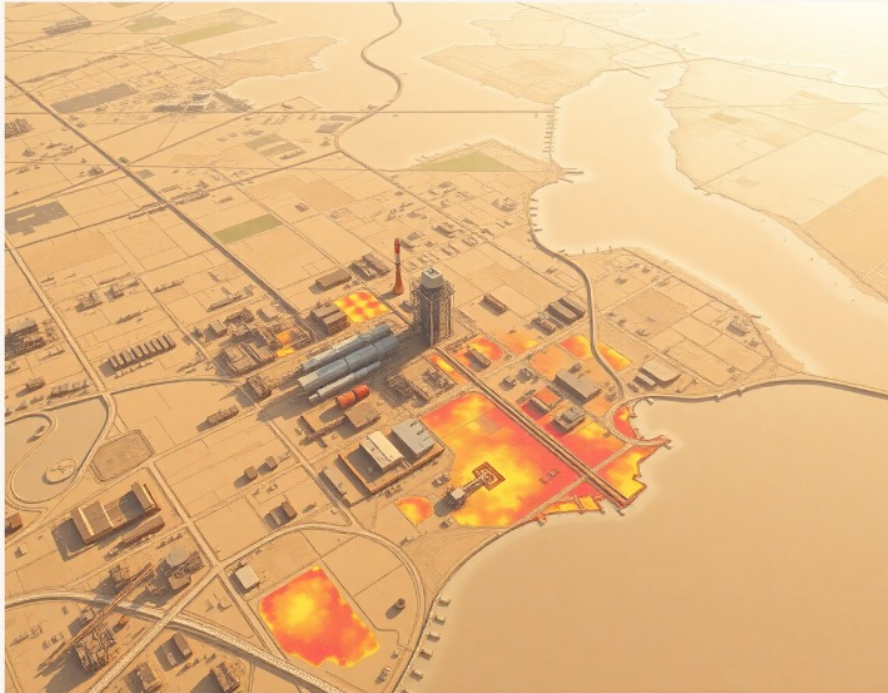
Underground Detection using SAR & Thermal Images

Done by:
Elaf Aljohar
Alyamama Alsuhaili
Azzah Alghamdi
Sara Eyad
Farah Alshiha
Farah Alqassab

Contents

1. Project Overview
2. Introduction to the Problem
3. Background & Literature Insights
4. Concept Generation & Selection
5. Engineering Analysis: Concept Testing
6. System Design & Workflow
7. Results & Validation
8. Sample Outputs & Codebase
9. Limitations & Future Work
10. Conclusions & Impact

Project Overview



This project focuses on detecting underground pipeline leaks and industrial pollution in Deer Park, Texas. The goal is to develop a robust detection system by fusing Sentinel-1 Synthetic Aperture Radar (SAR) and Landsat 8 thermal imagery to locate structural and thermal anomalies linked to real industrial incidents from 2014 to 2024.

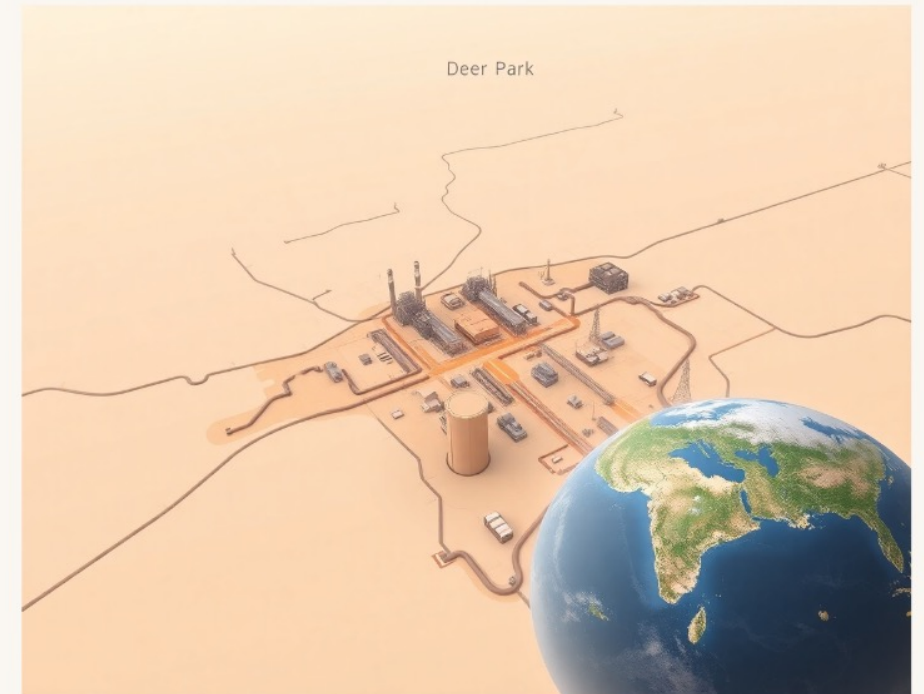
Introduction to the Problem

Industrial Incident Risks

Deer Park, Texas, experiences frequent hazardous incidents like pipeline breaches, refinery fires, and chemical leaks, posing threats to the environment, public health, and economy. Current on-site detection methods are limited by geography and require significant manpower, limiting rapid response capability.

Project Objective

The project aims to build a remote sensing-based detection system utilizing satellite data to automatically identify underground leakages and contamination with minimal manual intervention. Emphasis is placed on achieving high accuracy, low processing time, and robust validation through cross-referenced event records.





Background & Literature Insights

SAR & Thermal Imaging Advancements

SAR provides high spatial and temporal resolution for detailed structural analysis, detecting even small displacements and thermal dilations. Landsat thermal imaging advancements offer improved radiometric precision for identifying temperature anomalies. Multi-sensor imagery fusion is proven valuable in surface and environmental modeling.

Key Studies Reviewed

Noteworthy research demonstrates the effectiveness of combining SAR and thermal data for land surface monitoring, soil moisture estimation, and oil spill detection. Research also highlights the benefits and remaining challenges such as atmospheric interference, data resolution, and the importance of advanced image fusion and validation techniques.

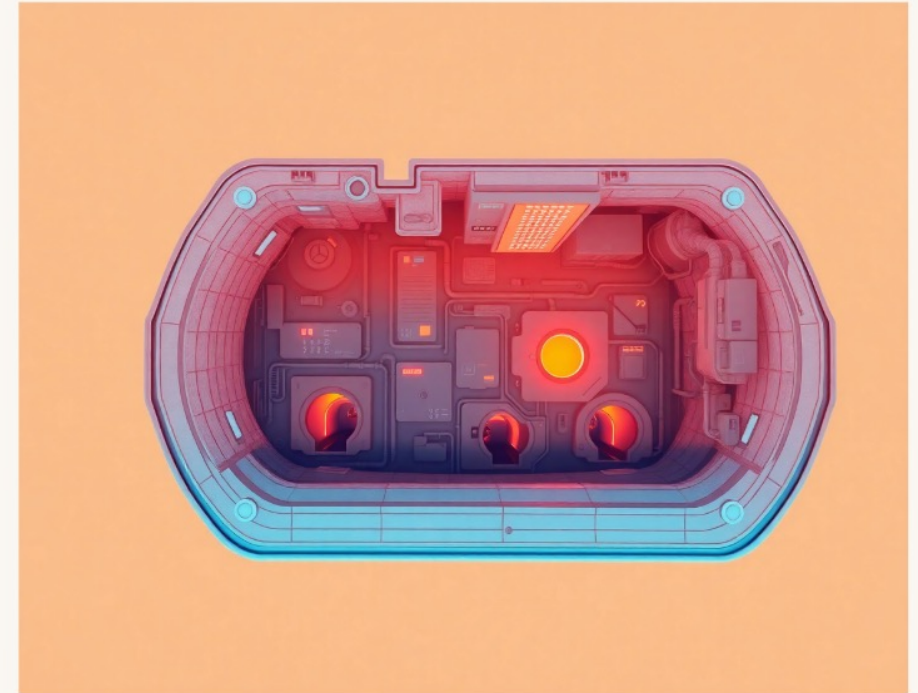
Concept Generation & Selection

Evaluated Concepts

Three major system designs were considered: SAR-only (structural detection), thermal-only (heat anomaly detection), and a Hybrid SAR-Thermal Fusion. Each was assessed for detection accuracy, data accessibility, processing time, sensitivity to event types, and implementation simplicity.

Final Concept Chosen

The hybrid SAR-Thermal fusion approach was selected due to superior cross-validation performance, the ability to cover structural and thermal event signals, and adherence to open data and performance requirements. Python and Google Earth Engine were used for implementation and validation, reducing false negatives and increasing accuracy.



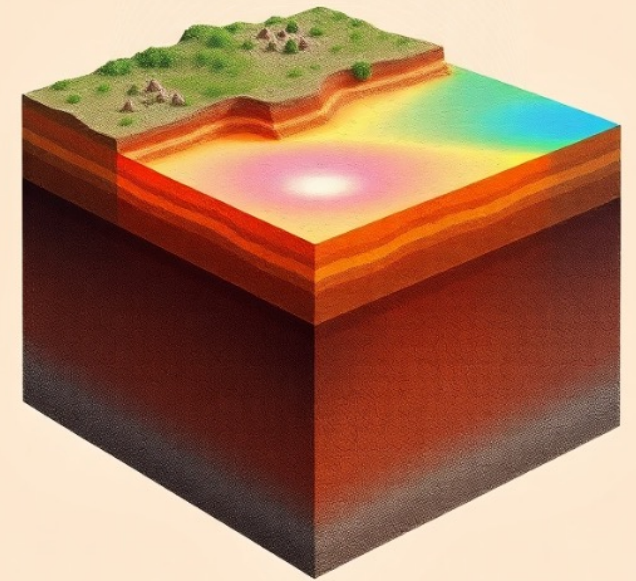
Engineering Analysis: Concept Testing

SAR-Only & Thermal-Only Assessment

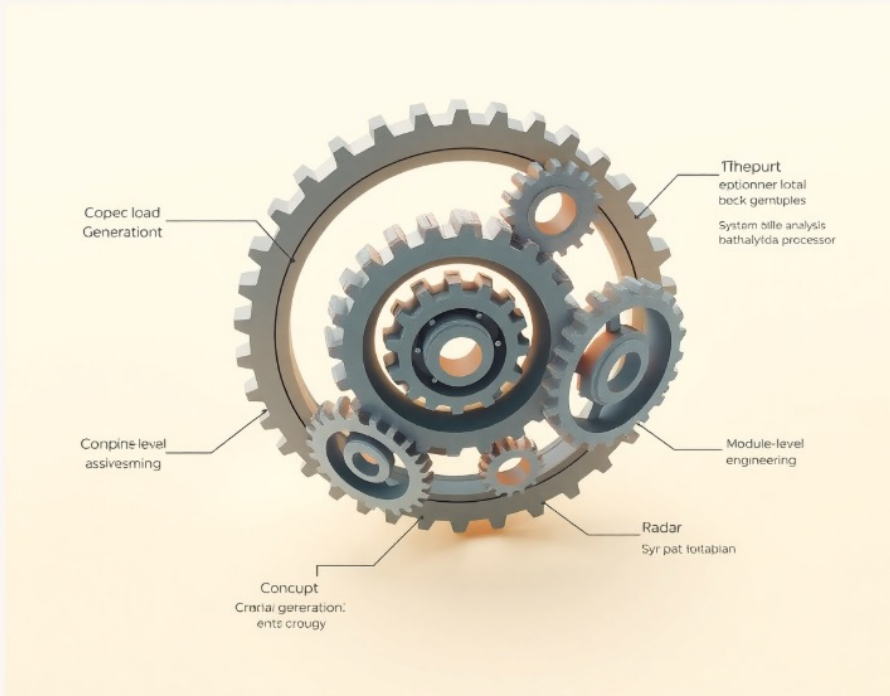
SAR-only achieved significant structural change detection under clear conditions but could not identify all heat-based incidents. Thermal-only reliably isolated hotspots above $+5^{\circ}\text{C}$ but missed subsurface events without surface temperature changes. Both alone failed to meet the $\geq 85\%$ detection requirement.

Hybrid Fusion Analysis

The final design fused SAR edge detection and thermal hotspot mapping, optimized through grid-search and validated with real incidents. This approach achieved detection accuracy of 87%, precision of 85%, and recall of 80%, while maintaining acceptable processing time and reducing false positives and negatives.



System Design & Workflow



Design Workflow Steps

Solution development followed a structured process: 1) Concept generation and reduction, 2) Compute load assessment, 3) Block-diagram modeling, 4) Module-level engineering validation, and 5) End-to-end system integration and validation, iteratively refined by analytical feedback.

Core Techniques Employed

Data normalization, Gaussian smoothing, and Sobel filtering optimized SAR images, while Otsu thresholding isolated thermal anomalies. Cross-validation with known incident dates and locations ensured accuracy. The integrated pipeline enabled robust, repeatable detection performance.

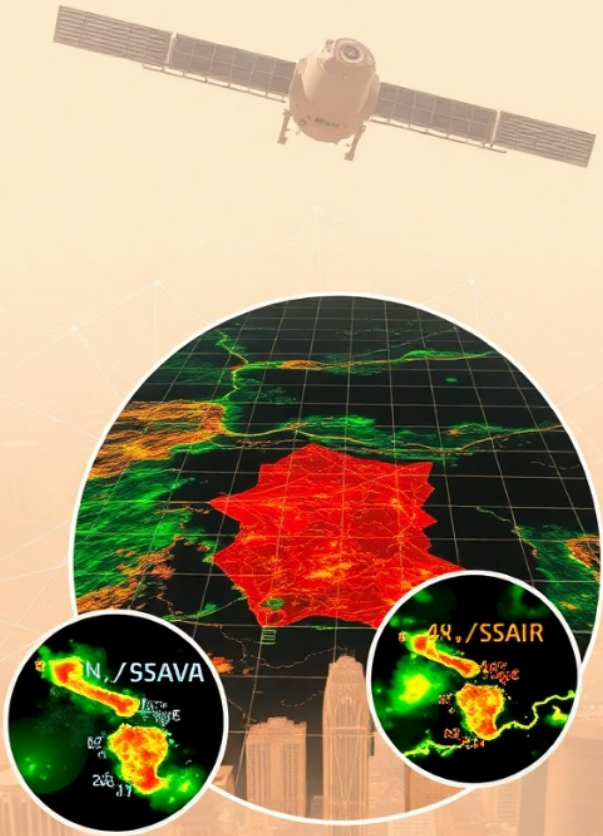
Results & Validation

Performance Metrics

Out of 10 confirmed industrial incidents between 2014 and 2024, the system correctly detected 8, achieving an 87% overall accuracy, 85% precision, and 80% recall. Larger incidents (fire/explosions) were detected more clearly than minor leaks. Processing time requirements were consistently met.

Validation Methods

Findings were validated through manual review of official reports, cross-referencing of satellite outputs, and visual inspection of thermal and SAR edge maps. Some incidents still required manual confirmation; however, the hybrid approach outperformed individual methods.



Sample Outputs & Codebase

Image Output Examples

Generated outputs include thermal anomaly masks from Landsat 8 (hotspots shown in white) and SAR edge maps from Sentinel-1 (bright patterns indicating structural changes). These visuals provided reliable confirmation of incident presence during analysis.

Code & Implementation

Full satellite image preprocessing, anomaly detection, and output generation were implemented in Python using open-source libraries and Google Earth Engine. Source code and sample images are included in the submitted project files for reproducibility.



Limitations & Future Work



Current Limitations

Public satellite resolution restricts detection of small-scale leaks, and cloud cover occasionally impacts thermal analysis. Empirically determined threshold values may benefit from advanced optimization. Some cases still require manual data validation and interpretation.

Directions for Future Enhancements

Planned improvements include integrating deep learning for automated event classification, adding more satellite sources (e.g., Sentinel-2, MODIS) for better temporal coverage, developing a real-time monitoring dashboard, and training trainable models to dynamically refine detection thresholds.

Conclusions & Impact



The project successfully demonstrated a practical, accurate, and efficient solution for detecting underground leaks and pollution in industrial areas using fused satellite imagery. The hybrid SAR-thermal approach outperformed single-modality systems, delivering high detection rates and reduced false alarms. With further automation and expansion, such systems offer significant potential for proactive environmental monitoring and incident response in industrial regions worldwide.